2026 SOURCE CODE ANALYSIS

Breaking the Algorithm

Growth Guide for Small X Accounts

Based on xai-org/x-algorithm (Jan 2026)

NO 2023 CONTAMINATION

The previous `twitter/the-algorithm` (TweepCred, SimClusters, Real Graph, MaskNet) has been **completely replaced**. None of those systems exist in the current algorithm. Every claim in this guide is from the 2026 source code or 2025-2026 empirical data.

Evidence tags: CODE from source code EMPIRICAL from published research INFERRED logical deduction

19	128	0
SCORING SIGNALS	HISTORY POSITIONS	HAND-ENGINEERED FEATURES

March 2026 · xai-org/x-algorithm · Rust + Python/JAX

What the Algorithm Sees Right Now

YOUR ACCOUNT: @longevityboris

96 followers · 104 following · Account since 2023 · Dormant · Free tier · Longevity/biotech niche

The Cold Start Problem CODE

The 2026 algorithm has **no** TweepCred, **no** SimClusters, **no** Real Graph. The README states:

> *"We have eliminated every single hand-engineered feature."*
>
> source-2026/README.md

Instead, a Grok-based transformer (`PhoenixModel`) learns everything from your last **128 engagement actions**:

```
history_seq_len: int = 128  // recsys_model.py
candidate_seq_len: int = 32
```

A dormant account has **zero** engagement history. The `UserActionSeqQueryHydrator` checks this:

```
if thrift_user_actions.is_empty() {
    return Err("No user actions found for user {user_id}");
}
```

TRANSLATION

If you have no recent engagement history, the Phoenix transformer **cannot score posts for you or about you**. The entire scoring pipeline short-circuits. Your posts cannot be properly ranked for anyone's feed. **Step zero is generating engagement history by actively using X.**

The 19 Scoring Signals CODE

From `weighted_scorer.rs` — every post is scored on exactly 19 signals:

SIGNAL	EST. WEIGHT	EVIDENCE
Follow Author	30x	INFERRED New in 2026. Ultimate discovery signal.
Share via DM	25x	INFERRED New in 2026. Separate dedicated signal.
Share via Copy Link	20x	INFERRED New in 2026. Off-platform sharing.
Reply	20x	EMPIRICAL Still high, but "reply_engaged_by_author" is GONE.

SIGNAL	EST. WEIGHT	EVIDENCE
Quote Tweet	18x	EMPIRICAL New dedicated signal in 2026.
Profile Click	12x	EMPIRICAL Consistent with 2023 empirical data.
Click (conversation)	10x	EMPIRICAL Deep interest signal.
Share (generic)	10x	INFERRED Opening share menu, before choosing channel.
Dwell (binary)	8x	EMPIRICAL Pausing on the post (2+ seconds).
Retweet	3x	EMPIRICAL Lower weight, lower probability.
Favorite (Like)	1x	EMPIRICAL Baseline. Most common, lowest weight.
Photo Expand	2x	INFERRED Moderate interest, image-specific.
Video Quality View	3x	CODE Gated behind MIN_VIDEO_DURATION_MS.
Dwell Time (continuous)	0.1/sec	CODE Only non-probability signal.
Quoted Click	4x	INFERRED Secondary engagement.
Not Interested	–20x	INFERRED Lightest negative. Content mismatch.
Mute Author	–40x	INFERRED Soft rejection.
Block Author	–74x	INFERRED Hard rejection.
Report	**–369x**	INFERRED Nuclear option. Devastating.

Weights are ESTIMATES. True values are in unpublished `params.rs`. Relative ordering is high-confidence.

What’s Dead from 2023

STOP CITING THESE — THEY'RE FROM DEAD CODE

2023 MYTH	2026 REALITY
TweepCred < 65 = only 3 tweets scored	**Eliminated**. No tweet limit.
Bookmarks = 10x likes	**NOT in weighted_scorer.rs**. Not a signal.
Reply + author engages back = 150x	**Gone**. reply_engaged_by_author removed.
Report = –369.0 exact weight	Report weight exists but value unknown.
SimClusters community embedding	**Replaced** by Phoenix two-tower retrieval.
Real Graph relationship scoring	**Replaced** by Grok engagement sequences.
Blue verified 4x / 2x boost	**Not in this repo**. Operates elsewhere.
Following/follower ratio penalty	**No ratio penalty in 2026 code**.

Step 1 — Wake the Account (Days 1–14)

GOAL: Fill your 128-position engagement history buffer

CODE The Grok transformer needs engagement data to work. Your `user_action_sequence` must have content before scoring works properly. A dormant account is algorithmically invisible.

Daily Routine (45–60 min)

Morning (20 min) — Engage Others

1. **Like 20–30 posts** in your niche. CODE This populates `history_actions` in the engagement sequence. The retrieval model learns your topics from these.
2. **Reply to 5–10 posts** from accounts with 1K–50K followers. Target posts under 30 minutes old. EMPIRICAL The 70/30 strategy: one creator grew 500→12,000 in 6 months.
3. **Quote 2–3 posts** with added context. CODE `quote_score` is a separate signal from `retweet_score` in `weighted_scorer.rs`.

Midday (15 min) — Create

4. **Post 1–2 original tweets**. Text-only with strong opening line. EMPIRICAL Buffer's 45M+ post analysis: text outperforms video by 30% on X.
5. **Reply to every reply** you receive within 30 min. CODE Reply chains fire `ServerTweetReply` for multiple participants.

Evening (10 min) — Network

6. **DM 1–2 people** a post you found valuable. CODE `share_via_dm_score` is a separate high-value signal — most underrated action in the algorithm.
7. **Follow 5–10 relevant accounts**. CODE Your following list determines what Thunder serves as in-network candidates.

Step 2 — Content Strategy (Days 15–60)

Format Rules

DO	DON'T	REASON
Text + opinion	Bare link	EMPIRICAL Link penalty largely removed Oct 2025, but engagement still lower
1–2 niche hashtags	5+ hashtags	EMPIRICAL 5+ hashtags → 40% reduction (Grok associates with spam)
Images users tap to expand	Stock photos	CODE `photo_expand_score` fires on tap — design for expansion
Videos > MIN_DURATION	Short clips	CODE `vqv_weight_eligibility()` zeroes VQV for short videos
Threads (5–7 tweets)	10 rapid single tweets	CODE `dwell_time` is continuous — threads maximize it
Space posts 2+ hours apart	Dump 5 posts in 1 hour	CODE `AuthorDiversityScorer` applies exponential decay

The Author Diversity Penalty CODE

```
fn multiplier(&self, position: usize) -> f64 {
    (1.0 - self.floor) * self.decay_factor.powf(position as f64) + self.floor
}
```

Your 1st post in a feed session gets full score. The 2nd gets `decay¹`, 3rd gets `decay²`. Space posts out.

DM Shares — The Hidden Weapon CODE

`share_via_dm_score` and `share_via_copy_link_score` are **separate dedicated signals** in `weighted_scorer.rs`. These are brand new in 2026. Most people ignore them entirely.

Action: When you see a great post in your niche, share it via DM to someone who'd appreciate it. When others DM your posts, it fires one of the highest-value signals in the algorithm.

Step 3 — How Discovery Works

Two Paths to Someone's Feed CODE

IN-NETWORK (Thunder)

Posts from accounts they follow. Sub-millisecond lookups from in-memory store. Your 96 followers see your posts here automatically.

Source: `thunder_source.rs`

OUT-OF-NETWORK (Phoenix)

ML-discovered posts via two-tower retrieval. Dot product similarity across global corpus. This is your growth engine.

Source: `phoenix_source.rs`, `recsys_retrieval_model.py`

The OON Penalty CODE

```
let updated_score = match c.in_network {
    Some(false) => base_score * p::OON_WEIGHT_FACTOR,
    _ => base_score,
};
```

Out-of-network posts get multiplied by `OON_WEIGHT_FACTOR` (< 1.0). Your posts reaching non-followers face a multiplicative penalty.

Candidate Isolation CODE

Each candidate post is scored **independently** — candidates cannot attend to each other in the transformer. Your score doesn't depend on which other posts are in the same batch. You don't "compete" with viral posts.

```
Candidates → Candidates: CANNOT attend (zeroed out)
Candidates → User/History: CAN attend (full attention)
```

How the Retrieval Model Finds You CODE

```
scores = jnp.matmul(user_representation, corpus_embeddings.T)
```

If users who engage with longevity content **also engage with YOUR content**, your posts get retrieved for other longevity-interested users. Engagement from niche-relevant accounts is what places you in the right embedding neighborhood.

What to NEVER Do

The Asymmetric Penalty System CODE

Negative scores are **compressed but devastating**:

```
fn offset_score(combined_score: f64) -> f64 {
    if combined_score < 0.0 {
        (combined_score + NEGATIVE_WEIGHTS_SUM) / WEIGHTS_SUM
            * NEGATIVE_SCORES_OFFSET
    } else {
        combined_score + NEGATIVE_SCORES_OFFSET
    }
}
```

- **Positive** scores scale linearly — more engagement = proportionally higher
- **Negative** scores are compressed into a narrow band near zero
- Even **moderate** negative predictions can drop your score below zero → compression branch → effectively dead

The Penalty Hierarchy

RANK	PENALTY	REVERSIBILITY
1	**VF safety drop** — post removed entirely (spam/violence)	Irreversible
2	**Block/mute filter** — removed before scoring happens	Unblock/unmute
3	**Report prediction** — P(report) drags score to zero	Model must relearn
4	**Block prediction** — P(block) weighted heavily	Model must relearn
5	**Mute prediction** — P(mute) weighted moderately	Model must relearn
6	**Not Interested prediction** — lightest negative signal	Model must relearn
7	**OON penalty** — out-of-network multiplier	Follow to bypass
8	**Author diversity decay** — repeated author in feed	Resets each session
9	**Rate-limit shadowban** — account-level suppression	48hrs – 3 months

CRITICAL: PREDICTIONS, NOT EVENTS

CODE The Grok transformer **predicts** that users would block/mute/report your content — and penalizes it **before anyone actually does**. Your past blocks and reports become training data that generalizes to your entire audience segment. One round of negative signals poisons future predictions.

When to Get Premium

What the Code Shows CODE

The Premium boost is **NOT in this repo** — not among the 19 signals in `weighted_scorer.rs`.

What Empirical Data Shows EMPIRICAL

- Buffer (18.8M posts): Premium accounts get 10x more reach
- Premium: 600 impressions/post vs. significantly lower for free
- Premium+: 1,550 impressions/post

Decision Framework

CONDITION	SUBSCRIBE?	WHY
Days 1–14	**NO**	Boost multiplies your score. Score × 0 = 0.
Day 21–30, posting consistently	**YES**	You now have engagement history. Boost has something to amplify.
You post links regularly	**YES**	Non-Premium link posts historically suppressed.

Milestones & Timeline

MILESTONE	EXPECTED	WHAT UNLOCKS
Engagement history populated	Day 7–14	Phoenix transformer can score your content
For You shows niche content	Day 7–10	Retrieval model learned your embedding
First reply-back from target	Day 2–3	Engagement edge established
Subscribe to Premium	Day 21–30	10x reach amplification
250 followers	Week 3–4	Meaningful in-network audience
1 post exceeding 1K impressions	Week 3–4	Out-of-network retrieval working
500 followers	**Month 2–3**	**Compounding growth begins**
First viral post (10K+ impressions)	Month 2–3	Embedding neighborhood established
1,000 followers	**Month 4–6**	**Self-sustaining flywheel**

Daily Checklist

MORNING (20 min)

- Like 20–30 niche posts (builds engagement history)
- Reply to 5–10 posts from larger accounts (< 15 min old)
- Quote 2–3 valuable posts with added context

MIDDAY (20 min)

- Post 1–2 original posts (spaced 2+ hours apart)
- Reply to ALL replies on your content within 30 min
- DM 1 great post to someone who'd value it

EVENING (10 min)

- Post 1 more original post or thread segment
- Check metrics: which posts got profile clicks? Double down.
- Follow 3–5 new relevant accounts

WEEKLY

- 1 thread (5–7 tweets) for dwell time
- 1 image post designed for tap-to-expand
- Review: which formats got most engagement?

The Scoring Formula

CODE From `weighted_scorer.rs`:

```
score = SUM(weight_i × P(action_i))
```

Where `P(action_i)` comes from the Grok transformer as `exp(log_probability)`.

```
Weighted Score =
  P(follow_author) ×  30
 + P(share_dm) ×  25
 + P(reply) ×  20
 + P(share_link) ×  20
 + P(quote) ×  18
 + P(profile_click) ×  12
 + P(click) ×  10
 + P(share) ×  10
 + P(dwell) ×  8
 + P(retweet) ×  3
 + P(vqv) ×  3 (if video > MIN_DURATION)
 + P(photo_expand) ×  2
 + P(favorite) ×  1
 + dwell_time ×  0.1 (seconds, not probability)
 + P(not_interested) ×  -20
 + P(mute) ×  -40
 + P(block) ×  -74
 + P(report) ×  -369
```

Weights are ESTIMATES based on code structure, 2023 baselines where applicable, and empirical 2025-2026 data. True values in unpublished `params.rs`.

The Full Pipeline CODE

1. **Query Hydration** — fetch engagement history + following list
2. **Candidate Sourcing** — Thunder (in-network) + Phoenix Retrieval (out-of-network)
3. **Hydration** — core data, author info, video duration, subscription status
4. **10 Pre-Scoring Filters** — dupes, age, self, muted keywords, blocked authors
5. **PhoenixScorer** — Grok transformer predicts P(action) for 19 actions
6. **WeightedScorer** — weighted sum + offset_score() compression
7. **AuthorDiversityScorer** — exponential decay for repeated authors
8. **OONScorer** — discount factor for out-of-network
9. **TopK Selection** — sort by score, take top K
10. **Post-Selection** — VFFilter (safety) + DedupConversationFilter

Source Code References

`weighted_scorer.rs`	19-signal scoring formula, offset_score()
`phoenix_scorer.rs`	Grok predictions → PhoenixScores extraction
`oon_scorer.rs`	Out-of-network weight factor
`author_diversity_scorer.rs`	Exponential decay for repeated authors
`recsys_model.py`	PhoenixModel: Grok transformer ranking
`recsys_retrieval_model.py`	Two-tower retrieval, ANN search
`grok.py`	Transformer + candidate isolation masking
`phoenix_candidate_pipeline.rs`	Full pipeline assembly + filter chain
`user_action_seq_query_hydrator.rs`	Engagement history fetch + empty check
`thunder_source.rs`	In-memory in-network post store

Source: xai-org/x-algorithm (Apache 2.0, January 2026)
Every claim tagged CODE EMPIRICAL or INFERRED
Full analysis: `docs/x-algorithm-analysis/`